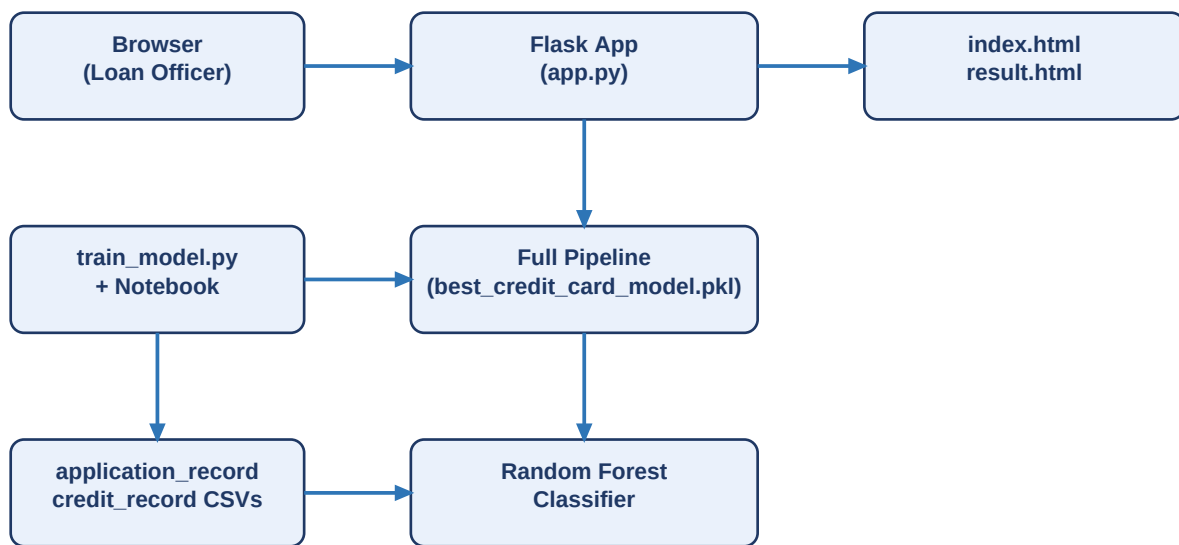


System Design Diagram

Credit Card Approval Prediction System | Project Design Phase

High-Level Architecture

Two paths: (a) **Offline Training** — `train_model.py` merges the two CSVs, derives the TARGET label from credit STATUS codes, trains four candidate models (Logistic Regression, Decision Tree, Random Forest, XGBoost), and saves the best as a full scikit-learn Pipeline (preprocessing + classifier) in `best_credit_card_model.pkl`. (b) **Online Serving** — Flask app loads the Pipeline once at startup and serves predictions through POST `/predict`.



Data Flow

- User fills the 17-field form on `index.html` and submits to POST `/predict`.
- JavaScript converts friendly age/experience (years) to `DAYS_BIRTH` / `DAYS_EMPLOYED` (negative integers) before submission.
- `app.py` reads raw form values, builds a pandas DataFrame in `FEATURE_COLUMNS` order, passes it to `pipeline.predict()`.
- The Pipeline internally applies `StandardScaler` to numeric columns and `OneHotEncoder` to categorical columns, then runs Random Forest inference.
- Prediction (0=Approved, 1=Rejected) is passed to `result.html` which renders a green-tick or red-cross result card with a risk message.

Pipeline Architecture (`train_model.py`)

Step	Component	Detail
1	ColumnTransformer	StandardScaler on numeric cols; OneHotEncoder(handle_unknown='ignore') on categorical
2	RandomForestClassifier	n_estimators=100, random_state=42 — best performer across Logistic Regression, Decision
3	joblib.dump	Saves full Pipeline to models/best_credit_card_model.pkl; saves column order to models/

Technology Stack

Layer	Technology / Version
Backend	Python 3.12.3, Flask 3.1.3, Werkzeug 3.1.8
ML Pipeline	scikit-learn 1.8.0 (ColumnTransformer, Pipeline, RandomForestClassifier)
Data	pandas 2.3.3, numpy 2.4.1, scipy 1.17.0
Model Serialization	joblib 1.5.3
Frontend	HTML5, CSS3 (dark theme, Plus Jakarta Sans font), Vanilla JS
Deployment	Gunicorn 26.0.0, Render (render.yaml + Procfile), Python 3.12.3